Exploratory Mapping of Buddhist Texts with a Three-Layer Map of Meaning, Style, and Source Markers: A Preliminary Analysis of Sectarian Reference Sets, Two Chinese Translations of the Amitabha Sutra, and Shinran's Writings

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Translation note. The authoritative edition of this paper is the Japanese print/book edition. This English version is an AI-assisted translation prepared for access by English-language readers. It preserves Chinese titles, Taisho identifiers, and key Japanese terms where needed, but wording and nuance should be checked against the Japanese edition for citation or scholarly use.

Abstract

This preliminary study proposes an exploratory framework for visualizing relations among Chinese Buddhist texts and Japanese Buddhist writings by separating textual proximity into three layers: a semantic layer, a stylistic/lexical layer, and a source-marker layer. The experiment uses 15 texts and 433 chunks, centered on reference texts associated with Japanese Buddhist traditions. Each text was embedded with `text-embedding-3-large`; a character n-gram TF-IDF layer was used as a stylistic/lexical contrast; and an explicit source-marker layer based on scripture titles, translator names, and fixed phrases was overlaid.

The two Chinese versions of the Amitabha Sutra are close in the semantic layer but distant in the stylistic/lexical layer. In the three-layer source-mixture map for *Kyogyoshinsho*（顯淨土眞實教行證文類）, the semantic, stylistic/lexical, and source-marker layers produce different source mixtures. This confirms that semantic proximity, stylistic/lexical proximity, and proximity as a clue to sources or learning paths are not the same. In the multilingual pilot, only the pair Toh 106 and T0676 was tested, and the nearest Chinese text to the 84000 English translation was the corresponding Chinese translation. The study remains preliminary: the corpus and marker dictionary are small, and the source-marker layer is not a complete detector of quotation or reference.

Keywords: Chinese Buddhist texts; Buddhist digital humanities; semantic embeddings; three-layer map; source markers; Amitabha Sutra; translation comparison; Shinran; Kyogyoshinsho; multilingual comparison

1 Introduction

1.1 Problem Setting

Japanese Buddhist traditions have historically emphasized particular scriptures, founder texts, and liturgical texts. Pure Land traditions, for example, center on the Three Pure Land Sutras and founder writings, while Shingon Buddhism places major esoteric scriptures such as the *Mahavairocana Sutra*, the *Vajrasekhara Sutra*, and the *Rishukyo* at the center. Zen traditions often cite the Heart Sutra, the Diamond Sutra, and the Vimalakirti Sutra, but their sectarian features may appear more clearly in records of sayings, monastic regulations, and founder writings than in sutras alone.

Recent large language models and embedding models can represent semantic proximity among texts as high-dimensional vectors. This study applies such methods to Buddhist texts and asks whether reference-text groups associated with Buddhist traditions can be placed on an exploratory map. It also uses different Chinese translations of the Amitabha Sutra and multiple texts by the same translators to examine how semantic proximity and translation style diverge. In addition, it visualizes, at the chunk level, why semantic proximity, stylistic proximity, and proximity as source or learning path cannot be treated as identical in Shinran's writings. A small comparison with an 84000 Reading Room English translation is included to test whether semantic embeddings can recover same-text relations across languages.

1.2 Related Work and Position of This Study

Digital resources such as SAT, CBETA, the 84000 Reading Room, and BDRC have made large-scale comparison, translation comparison, and parallel-passage exploration increasingly feasible in Buddhist studies[1, 2, 3, 10]. For Chinese Buddhist texts, stylometric studies have used character n-grams, particles, function words, PCA, and clustering to examine translator, period, and stylistic differences[24, 25]. Such work is connected both to broader authorship-attribution methods[21, 22] and to lexical resources for Chinese Buddhist translations[23].

Embedding methods have also already been applied to Buddhist materials. Huang and Wang investigate word embeddings for CBETA Chinese Buddhist texts, including evaluation data, chronological and translator-specific models, word-sense estimation, and extraction of core concepts[26]. Embedding evaluation has also been attempted for Buddhist Sanskrit[29]. Therefore, this paper does not claim novelty simply by applying embeddings to Buddhist texts.

Quotation, reference, and intertextuality have likewise been studied through computational methods. Nehrdich proposes a method for detecting parallel passages in Chinese Buddhist sources using continuous word representations and million-scale nearest-neighbor search[27], and DharmaNexus supports intertextuality exploration across Buddhist texts[30]. The chunk-neighbor mixing rate and bridge chunks used here do not replace large-scale parallel-passage detection. They are preliminary indicators for locating semantic proximity and partial overlap at the level of texts and chunks.

Multilingual Buddhist NLP is also an active area. Felbur, Meelen, and Vierthaler studied cross-linguistic semantic textual similarity between Buddhist Chinese and Classical Tibetan[28]. A recent preprint proposes MITRA, a multilingual corpus and pretrained language model for Pali, Sanskrit, Buddhist Chinese, and Tibetan[31]. The comparison between the 84000 English translation and Chinese *Samdhinirmocana-sutra* in this paper should therefore be read as a small pilot connected to this broader movement, not as the first demonstration of multilingual Buddhist retrieval.

The contribution of this paper lies elsewhere. It treats reference texts associated with Japanese Buddhist traditions not as labels for a classifier but as an exploratory map in semantic space. It combines text-level mean vectors, chunk distributions, chunk-neighbor mixing, translation comparison, and source/reference indicators for Shinran's writings into one preliminary framework. It further separates semantic proximity, stylistic/lexical proximity, and proximity based on explicit source markers such as scripture titles, translator names, and fixed phrases on the same chunk sequence. The central claim is that embeddings are useful for exploring thematic proximity, but sectarian tradition, translator style, and citation paths require separate layers of stylistic features and source-marker information.

The aim is not to build a correct classifier of Buddhist sects. Rather, it is to ask what kinds of proximity and interpretive problems appear when texts referenced by different traditions are placed in a shared embedding space.

2 Research Questions

This paper addresses five questions.

1. Can semantic embeddings recover thematic clusters such as the Three Pure Land Sutras or Shingon esoteric scriptures?
2. If reference-text groups for each tradition are visualized as centroids, how well do they represent the thematic proximity of the input texts?
3. When different Chinese translations of the Amitabha Sutra and multiple texts by the same translator are compared, can semantic proximity and translation style be distinguished?
4. How do chunks of *Kyogyoshinsho* differ across the semantic, stylistic/lexical, and source-marker layers in relation to the Three Pure Land Sutras and the two Amitabha Sutra translations?
5. When Chinese Buddhist texts and 84000 English translations are compared, does same-text identity appear as a cross-language nearest-neighbor relation?

3 Data

Most source texts were obtained from the SAT Taisho Shinshu Daizokyo Text Database. For *Kyogyoshinsho*（顯淨土眞實教行證文類）, the study used the *Jodo Shinshu Seiten* Seikyo Database of the Jodo Shinshu Hongwanji-ha Research Institute. Some verification texts were existing local acquisitions and may not be complete. The corpus used here consists of 15 texts and 433 chunks. The multilingual pilot uses the SAT Chinese *Samdhinirmocana-sutra*（解深密經, T0676）and the 84000 Reading Room English translation *The Teaching Explaining the Thought*（Toh 106）. The 84000 text is treated as a research cache, and the text itself is not redistributed here. Public outputs are limited to acquisition procedures, preprocessing scripts, chunk identifiers, similarity matrices, figures, and other derived information.

For the analysis of Amitabha Sutra references in Shinran's writings, a page of *Nyushutsu Nimonge*（入出二門偈）from the Shinshu Seiten search site was also consulted. This material was not included in the global sectarian mean-vector map; it was used only to inspect explicit scripture and translator names, phrase indicators, and nearest-neighbor relations between *Kyogyoshinsho* chunks and the two Amitabha Sutra translations.

Table1: Texts and Corpus Information

Label	Text	ID/Source	Coverage	Characters/Chunks	Notes
Pure Land	佛說無量壽經	T0360/SAT	v0 range	10307/27	Translation attributed to 康僧鎧; Larger Sutra.
Pure Land	佛說觀無量壽佛經	T0365/SAT	v0 range	9003/23	Translation attributed to 畺良耶舍; Contemplation Sutra.
Pure Land	佛說阿彌陀經	T0366/SAT	v0 range	2368/7	Kumarajiva translation; Smaller Sutra.
Pure Land	稱讃淨土佛攝受經	T0367/SAT	v0 range	4630/12	Xuanzang translation; comparison layer for the Amitabha Sutra.
Jodo Shinshu	顯淨土眞實教行證文類	Seikyo DB	acquired text	78978/197	Shinran's writing; Seikyo DB used.

Label	Text	ID/Source	Coverage	Characters/Chunks	Notes
Lotus	妙法蓮華經	T0262/local	excerpt	4718/13	Kumarajiva translation; provisional Lotus evaluation.
Lotus/Zen	觀世音菩薩普門品	T0262/SAT range	chapter	2134/6	Chapter 25, often read as the Kannon Sutra.
Shingon	大毘盧遮那成佛神變加持經	T0848/SAT	v0 range	10770/28	Mahavairocana Sutra.
Shingon	金剛頂一切如來眞實攝大乘現證大教王經	T0865/SAT	v0 range	8342/22	Representative Vajrasekhara text.
Shingon	大樂金剛不空眞實三麼耶經	T0243/SAT	v0 range	3309/9	Rishukyo.
Zen/Shingon	般若波羅蜜多心經	T0251/SAT	v0 range	1307/3	Xuanzang translation; liturgical layer.
Hosso	解深密經	T0676/SAT	v0 range	7082/18	Xuanzang translation; Chinese counterpart for multilingual pilot.
Zen	金剛般若波羅蜜經	T0235/SAT	v0 range	5909/15	Kumarajiva translation; Zen reference candidate.
Zen	維摩詰所説經	T0475/SAT	v0 range	12129/32	Kumarajiva translation; Zen reference candidate.
Huayan	大方廣佛華嚴經	T0279/SAT	v0 range	7743/21	Translation attributed to 實叉難陀; Eighty-fascicle Huayan.

Note: “v0 range” indicates the portion processed in this experiment and does not guarantee philological completeness. Translator names follow SAT and Taisho catalog conventions; T0365 is treated as 畺良耶舍 and T0279 as 實叉難陀.

This experiment is preliminary. Not all large scriptures were processed as complete texts. The “v0 range” in the table means the range successfully processed from SAT or local acquired files for this experiment. In particular, the Lotus Sutra sample is short and likely not a complete text. The purpose is to summarize experimental results and derived data, not to redistribute SAT source text.

4 Methods

4.1 Chunking and Embeddings

For SAT-derived texts, the script extracted text lines from HTML, removed display buttons, image references, and extra whitespace, and concatenated the remaining lines. For local acquired texts and the Seikyo DB text, non-empty lines were concatenated and whitespace removed. Punctuation, variant characters, and editorial marks were not rigorously normalized. The preprocessing is therefore sufficient for a first semantic-embedding experiment, but not for strict stylistic analysis.

The texts were tokenized with `tiktoken`'s `cl100k_base` encoding and chunked into 700-token windows with 100-token overlap. Natural boundaries such as fascicles, chapters, and paragraphs were not used in v0. Each chunk was embedded with `text-embedding-3-large`. Embeddings for the same model and same text chunk were cached to avoid repeated API calls. Terms such as model, tokenizer, chunk, and similarity are summarized in the appendix.

Text-level vectors were computed as simple averages of the chunk vectors belonging to each text. Explicit L2 normalization was not applied before averaging or after averaging. Cosine similarity was used at comparison time, so vector length was normalized during similarity calculation.

4.2 Similarity and Centroids

Text-to-text similarity was computed with cosine similarity. A sectarian reference-group centroid was defined as the mean of text vectors labeled `core_sutra` or `founder_text` for that tradition. Here `core_sutra` and `founder_text` are working labels for central scriptures and founder writings. Translator centroids were similarly defined as means of text vectors associated with the same translator. Because the number of texts per translator is small, translator centroids are used only as visual guides.

4.3 Three-Layer Scores and the Source-Mixture Map

For the two Amitabha Sutra translations and Shinran's writings, this paper separates three layers: the semantic layer, the stylistic/lexical layer, and the source-marker layer (Figure 1). A figure that overlays these layers on the same target chunk sequence is called a three-layer source-mixture map. The central proposition is that semantic proximity, stylistic/lexical proximity, and proximity as a clue to sources or learning paths are not the same.

For chunks a_i and b_j, three scores are defined. The semantic score S_{ij} is the cosine similarity of `text-embedding-3-large` chunk embeddings. The stylistic/lexical score T_{ij} is cosine similarity based on character 2-to-5-gram TF-IDF. The TF-IDF vectorizer was fit on the combined set of *Kyogyoshinsho* chunks and the four reference-source chunk sets. This layer is not strict stylometry; it is an initial operationalization of lexical and orthographic proximity. The source-marker score $C_i(s)$ counts occurrences of dictionary markers for each source s, including scripture titles, translator names, fixed phrases, terms characteristic of the two Amitabha translations, and phrases related to the Three Pure Land Sutras. The source-marker layer is an exploratory proxy based on explicit markers and is not a complete detector of quotation or reference.

For the *Kyogyoshinsho* source-mixture map, the reference sources were limited to the Larger Sutra, the Contemplation Sutra, Kumarajiva's Amitabha Sutra, and Xuanzang's *Praise of the Pure Land and the Buddha's Protective Acceptance Sutra*. Let C_s be the set of chunks belonging to source s. For target chunk i, the semantic and stylistic/lexical scores toward source s are $m_i^S(s) = \max_{j \in C_s} S_{ij}$ and $m_i^T(s) = \max_{j \in C_s} T_{ij}$. This maximum-similarity method finds the nearest candidate within each reference source; because sources with more chunks may more easily produce higher maxima, the result is treated as an exploratory relative indicator, not an estimate of absolute contribution.

Semantic and stylistic/lexical source weights were converted into relative weights among the four sources with softmax. The semantic layer used $\tau_S = 0.04$ and the stylistic/lexical layer used $\tau_T = 0.025$. For numerical stability the implementation subtracts each row maximum before exponentiation, but it does not standardize scores or use z-scores. These temperature parameters are preliminary settings chosen to make relative source differences visible; they were not optimized on independent validation data.

$$w_i^S(s) = \frac{\exp(m_i^S(s)/\tau_S)}{\sum_{s'} \exp(m_i^S(s')/\tau_S)}, \quad w_i^T(s) = \frac{\exp(m_i^T(s)/\tau_T)}{\sum_{s'} \exp(m_i^T(s')/\tau_T)}$$

In the source-marker layer, dictionary-marker counts were normalized by source. Chunks with no detected markers were assigned to "unmarked." When markers from multiple sources occurred in the same chunk, weights were divided in proportion to counts. The semantic and stylistic/lexical layers were displayed with a seven-chunk moving average, and the source-marker layer with a nine-chunk moving average. Weights are relative within each layer and should not be compared numerically across layers. Because the semantic and stylistic/lexical layers always distribute weight among the four sources, an "other" category for sources outside the four-source set is not yet included.

Table2: Representative Examples from the Source-Marker Dictionary

Source	Marker examples	Type
Larger Sutra	無量壽, 法藏, 本願, 四十八願, 願生, 安樂, 正覺	Title/doctrinal terms
Contemplation Sutra	觀無量壽, 韋提希, 頻婆娑羅, 十六觀, 日想觀, 水想觀, 九品	Title/proper names
Kumarajiva Amitabha	阿彌陀經, 阿彌陀, 舍利弗, 執持名號, 一心不亂, 恒河沙, 極樂	Title/fixed phrases
Xuanzang Praise Pure Land	稱讃浄土／稱讚淨土, 稱讃浄土經／稱讚淨土經, 玄奘, 舍利子, 恒河沙, 慈悲加祐, 攝受法門, 百千俱胝	Title/translator/fixed phrases

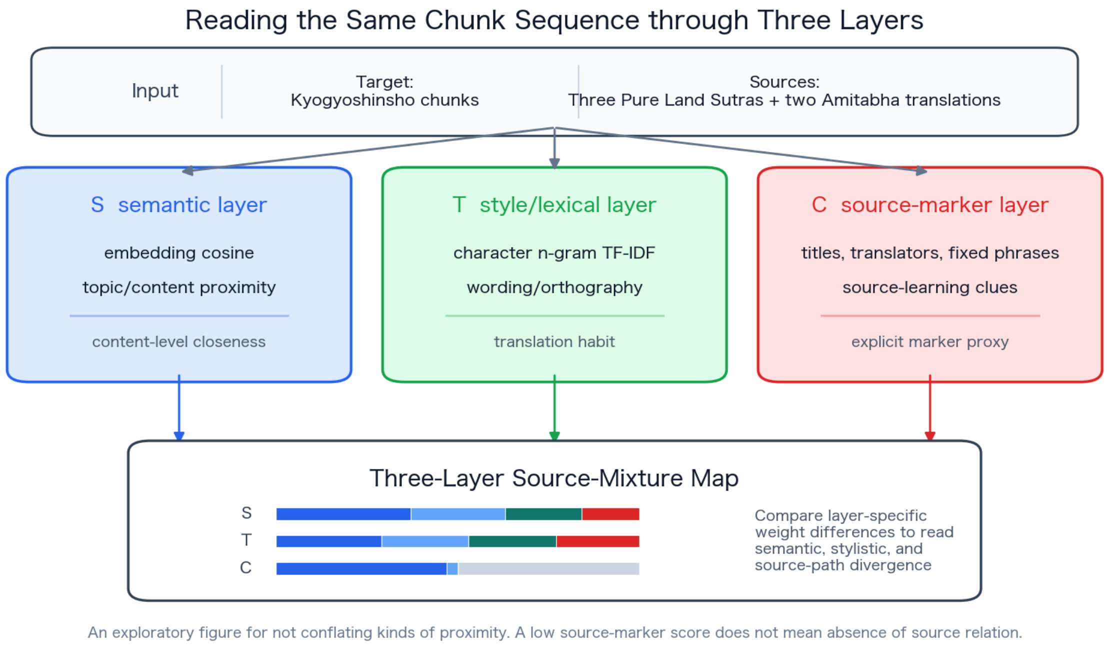

Figure1: Conceptual diagram of the three-layer source-mixture map. The same target chunk sequence is read through the semantic, stylistic/lexical, and source-marker layers, and the differences among layer-specific weights are interpreted as differences in kinds of proximity. The source-marker layer is not a complete detector of quotation or reference; it is an exploratory proxy based on explicit markers.

4.4 Visualization and Contrastive Experiments

Text vectors, sectarian reference-group centroids, and translator centroids were placed in a shared space and visualized in two dimensions with PCA. The PCA random seed was `random_state=0`. In the text-mean map, the explained variance of PC1 and PC2 was 23.3% and 18.7%, respectively, for a combined 42.0%. For the two Amitabha Sutra translations, character-level TF-IDF was also computed with character 2-to-5-grams, as a contrastive analysis of lexical and orthographic proximity rather than semantic proximity.

Mean vectors alone hide the internal spread of long scriptures and multi-topic writings. Therefore, major texts were also visualized as chunk-level embeddings with one-standard-deviation ellipses. In the main chunk-distribution figure, the explained variance of PC1 and PC2 was 6.9% and 4.4%, respectively, for a combined 11.4%.

Because overlap in a two-dimensional projection can be distorted, the chunk-neighbor mixing rate was computed in the original high-dimensional embedding space. For two texts A and B, let $C = A \cup B$ be the combined chunk set. For each chunk $c \in C$, the top-k neighbors $N_k(c)$ were found after excluding the chunk itself. The proportion of neighbors in $N_k(c)$ that belong to the other text is $r(c)$, and the mixing rate is $M(A, B) = |C|^{-1} \sum_{c \in C} r(c)$. Unless otherwise noted, $k = 5$.

4.5 Multilingual Pilot

The multilingual pilot extracted the 84000 Reading Room English translation *The Teaching Explaining the Thought*（Toh 106）from HTML and compared it with the SAT Chinese *Samdhinirmocana-sutra*（解深密經, T0676）. Chinese controls included the Diamond Sutra, Vimalakirti Sutra, Heart Sutra, and Amitabha Sutra. The evaluation used the nearest-neighbor rank of the corresponding Chinese text from the English text vector.

5 Results

5.1 Semantic Placement of Sectarian Reference Texts

Figure 2 shows the PCA map of text vectors and translator centroids. Pure Land reference texts tend to appear near one another in the upper-left direction, and Shingon esoteric scriptures tend to cluster downward. In contrast, Zen scriptures alone did not clearly separate Soto, Rinzai, and Obaku traditions. This does not mean that no sectarian differences exist; it means that the scripture layer used here does not adequately represent those traditions.

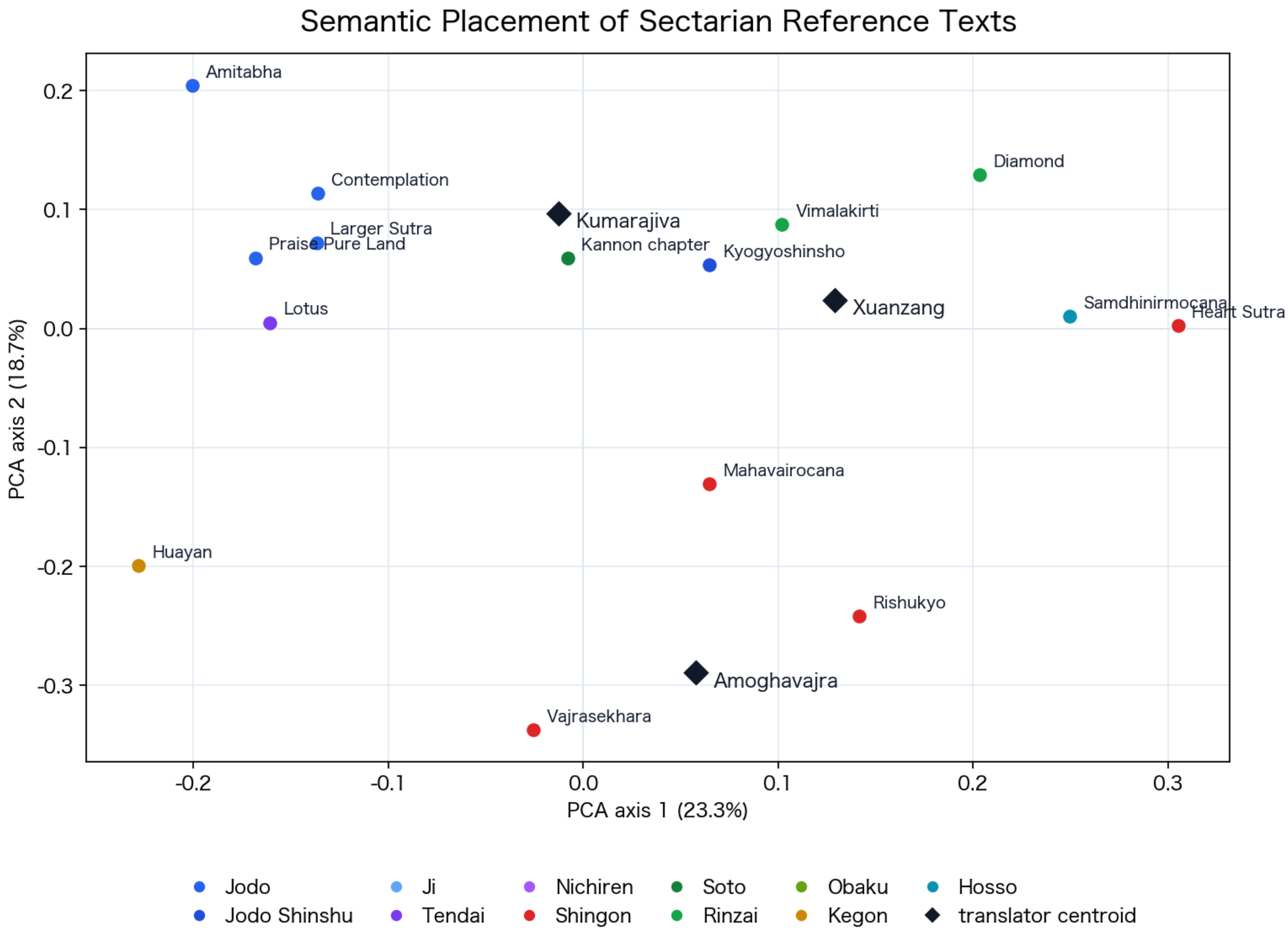

Figure2: Semantic placement of sectarian reference texts. Circles indicate texts and diamonds indicate translator centroids. The explained variance of the PCA axes is shown in the axis labels.

5.2 Chunk Distributions and Spread

Figure 3 visualizes chunks rather than text means. Ellipses show the one-standard-deviation range of each text's chunk group. Texts that become single points in a mean-vector map may have internal thematic spread or partial overlap with other texts. *Kyogyoshinsho*, which contains quotations and doctrinal synthesis, has a particularly wide chunk distribution.

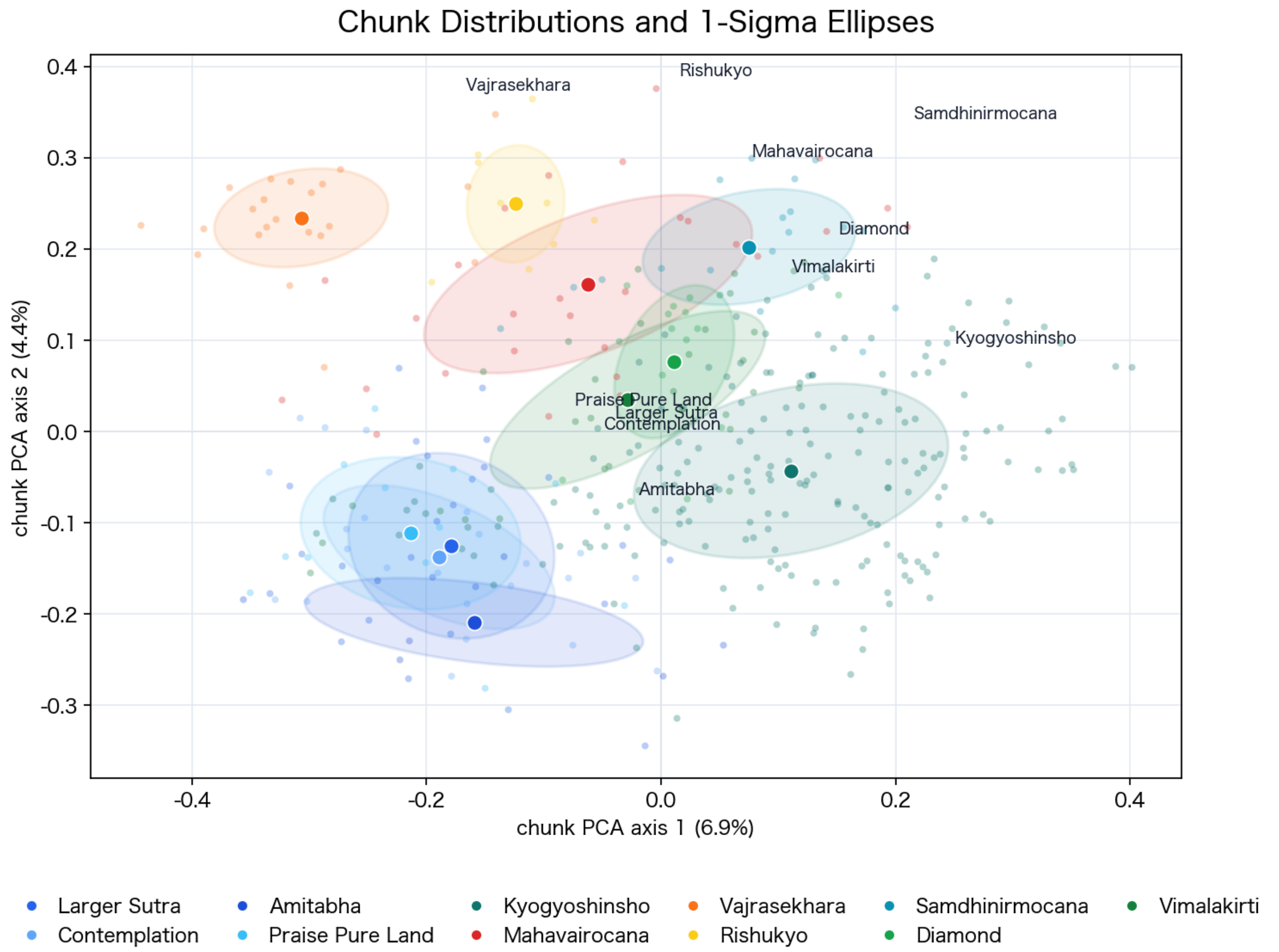

Figure3: Chunk distributions and one-standard-deviation ellipses for major texts. Small points are chunks and darker points are chunk centroids for each text. Because the explained variance of the two-dimensional PCA is low, overlap should be interpreted together with high-dimensional neighbor-mixing measures.

Figure 4 shows representative local PCA panels. The two Amitabha Sutra translations overlap not only in mean vectors but also in chunk distributions. Other groups, such as esoteric texts or Prajna-paramita/Hosso/Zen controls, show cases where mean proximity and distribution shape do not coincide.

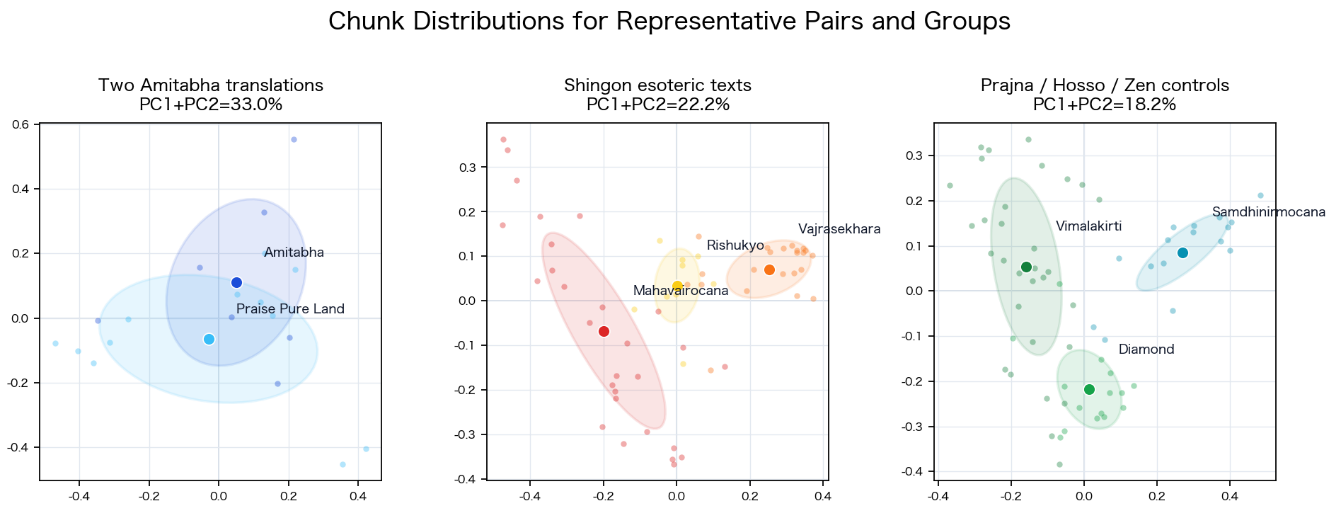

Figure4: Chunk distributions for representative pairs and groups. PCA is recalculated within each panel to make local overlap easier to see.

Figure 5 shows the high-dimensional chunk-neighbor mixing rate. A higher value means that chunks from the other text more often appear among a chunk's nearest neighbors. The two Amitabha Sutra translations have a high value of 0.37, indicating distributional mixing across translations. Pairs with larger thematic differences, such as the Amitabha Sutra and the Rishukyo, show low mixing.

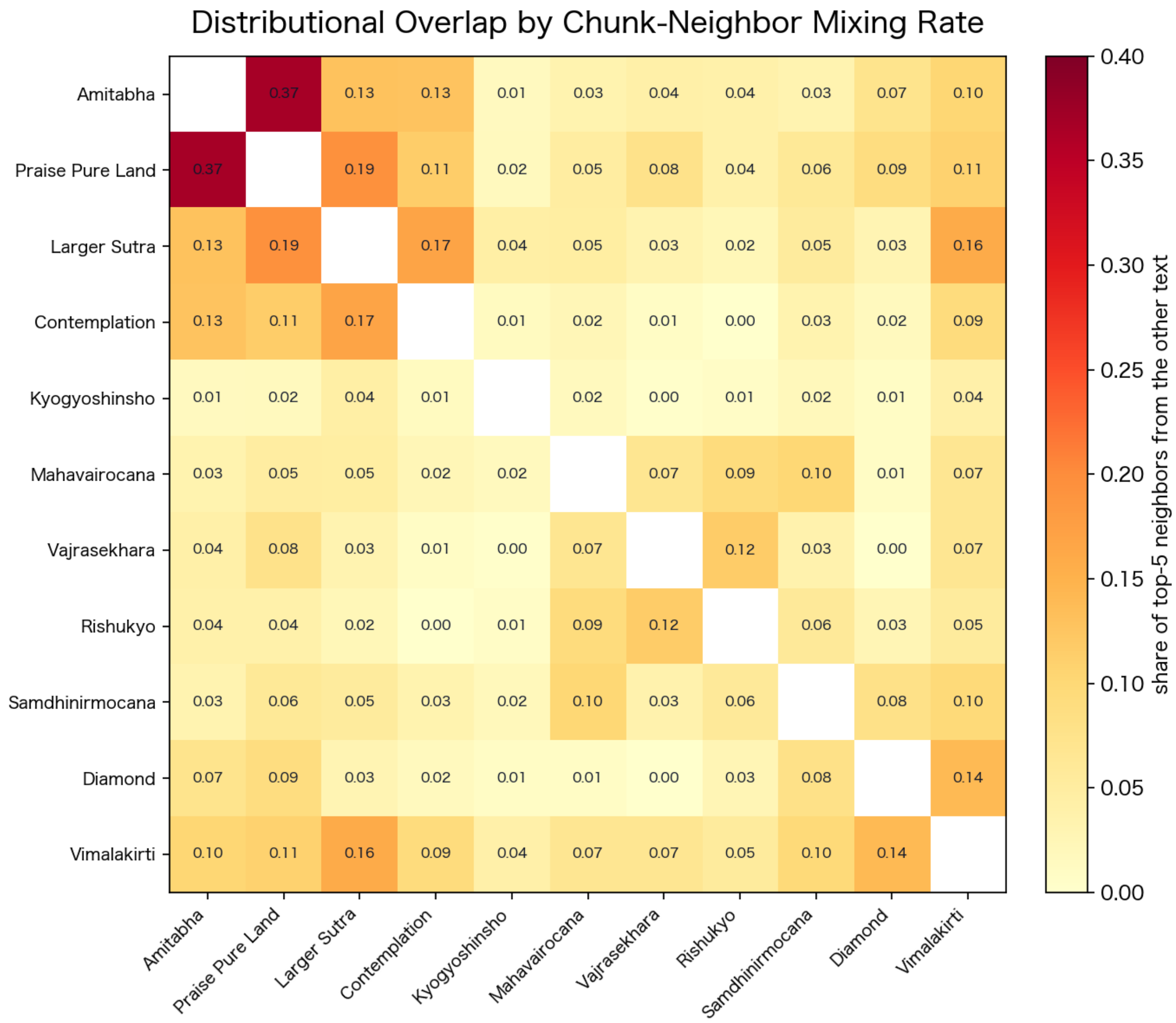

Figure5: Distributional overlap by chunk-neighbor mixing rate. Each cell is the average proportion of top-5 neighbors that belong to the other text when the chunks of the two texts are combined.

Figure 6 compares text-mean similarity and chunk-neighbor mixing. The two Amitabha Sutra translations appear in the upper-right: their mean vectors are close and their chunks mix strongly. *Kyogyoshinsho* and the Three Pure Land Sutras are close in mean-vector similarity, but their chunk-neighbor mixing is lower. This reflects the composite nature of *Kyogyoshinsho*: it is grounded in the Three Pure Land Sutras but contains quotations, exegesis, and doctrinal exposition, so not all chunks mix uniformly with sutra chunks.

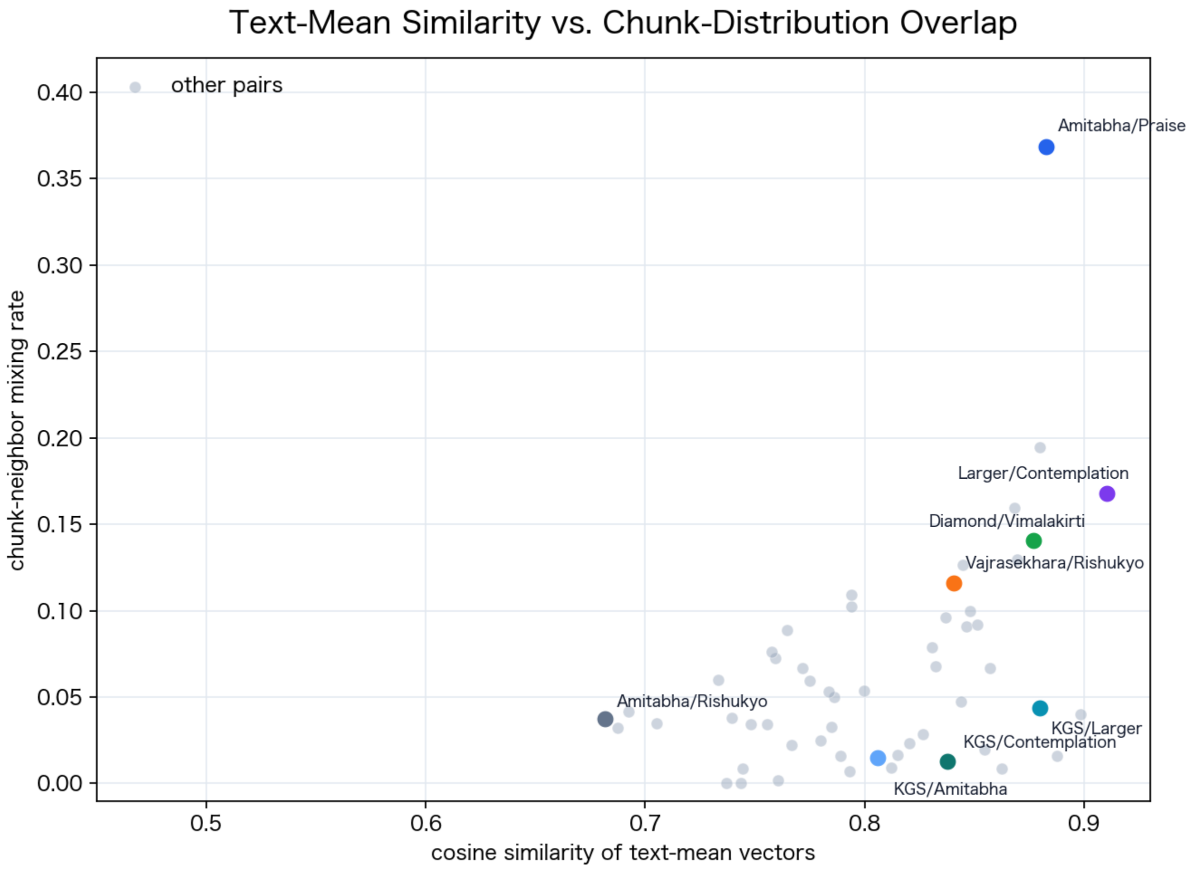

Figure6: Relation between text-mean vector similarity and chunk-neighbor mixing rate. Points farther to the upper right are close both as text means and as local chunk distributions.

Table3: Top-k Sensitivity of Chunk-Neighbor Mixing for the Two Amitabha Sutra Translations

top-k	Observed mixing	Rank among 55 pairs	Complete-mixing mean	95% range
1	0.1579	1	0.4897	0.2105–0.7368
3	0.2632	1	0.4911	0.3333–0.6140
5	0.3684	1	0.4911	0.3787–0.5789
10	0.4474	1	0.4908	0.4263–0.5421

Across $k = 1, 3, 5, 10$, the two Amitabha Sutra translations rank first among the 55 compared pairs. However, the label-randomized complete-mixing reference distribution has a higher mean than the observed values. This reference distribution should not be read as an ordinary significance test; it indicates what complete label mixing would look like if labels were independent of semantic structure. The main interpretation therefore rests on relative rank and top-k sensitivity among real text pairs.

Table4: Bridge Chunks for Representative Pairs. KGS denotes Kyogyoshinsho.

Pair	Nearest chunk pair	Chunk sim.	Text sim.	Mixing
Amitabha/Praise	T0366 0005 / T0367 0010	0.7672	0.8825	0.3684
KGS/Larger Sutra	KGS 0002 / T0360 0004	0.7969	0.8797	0.0438
KGS/Contemplation	KGS 0088 / T0365 0002	0.7095	0.8375	0.0127
KGS/Amitabha	KGS 0098 / T0366 0005	0.7250	0.8058	0.0147
Larger/Contemplation	T0360 0021 / T0365 0008	0.7660	0.9102	0.1680
Vajrasekhara/Rishukyo	T0865 0000 / T0243 0000	0.8176	0.8405	0.1161
Diamond/Vimalakirti	T0235 0004 / T0475 0022	0.7235	0.8770	0.1404
Amitabha/Rishukyo	T0366 0005 / T0243 0007	0.5900	0.6816	0.0375

For *Kyogyoshinsho* and the Three Pure Land Sutras, bridge-chunk similarities are high, roughly 0.71 to 0.80. Partial correspondence with the sutras is therefore visible, but the low mixing rate shows that the whole *Kyogyoshinsho* distribution does not simply become identical with the sutra distributions.

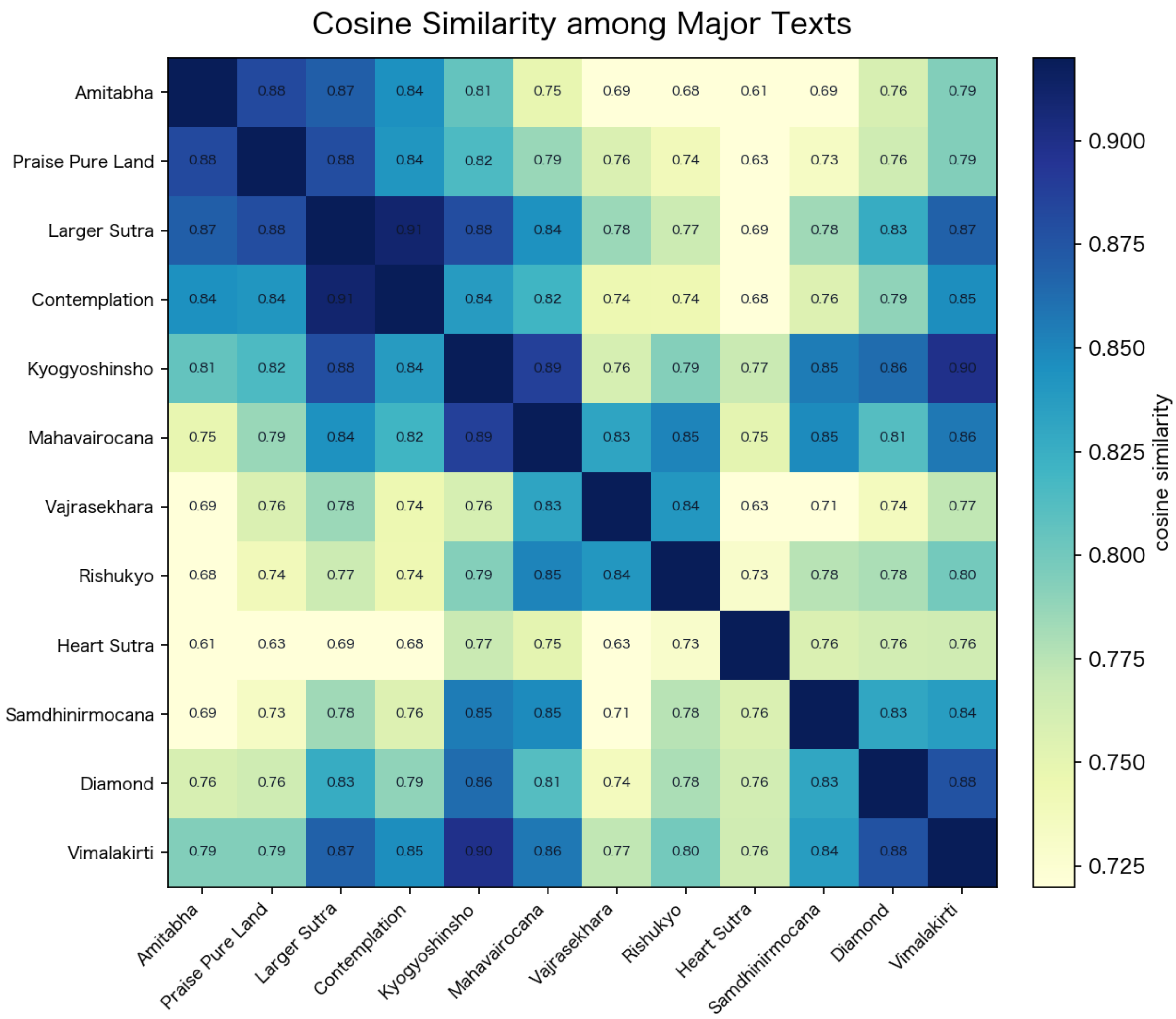

Figure7: Cosine similarities among major texts. Larger values indicate greater proximity in embedding space.

5.3 The Two Amitabha Sutra Translations

Kumarajiva's 佛説阿彌陀經（T0366）and Xuanzang's 稱讃淨土佛攝受經（T0367）show high similarity in semantic embeddings but low similarity in character-level TF-IDF (Figure 9). In the three-layer framework, they are a baseline case: they correspond strongly in the semantic layer while remaining distant in the stylistic/lexical layer.

Figure 8 recomputes three indicators for the two translations in the sectarian reference corpus. The text-mean vector similarity is high (0.8825). The character n-gram TF-IDF similarity for the same texts is only 0.1833. The chunk-neighbor mixing rate is 0.3684, the highest among the compared pairs. Thus, the two translations show same-content proximity in the semantic and distributional layers, while translation and orthographic differences remain visible in the stylistic/lexical layer.

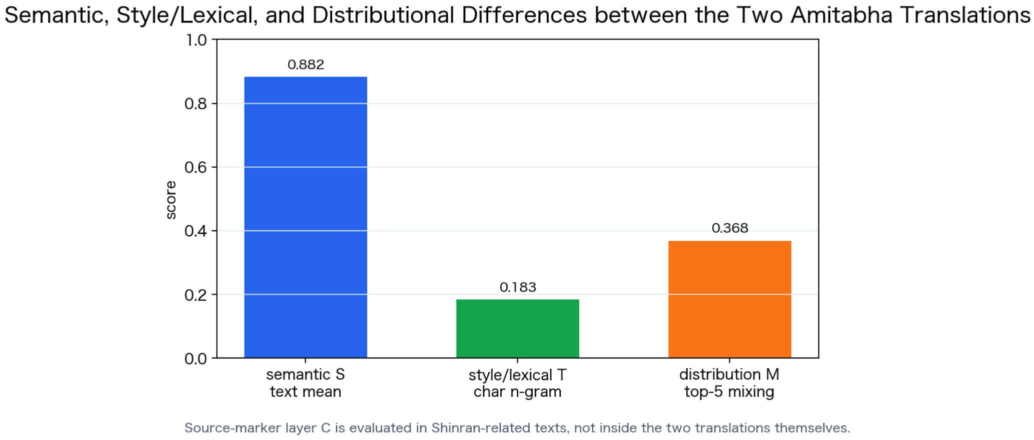

Figure8: Semantic, stylistic/lexical, and distributional differences between the two Amitabha Sutra translations. The source-marker layer is evaluated not within the two translations themselves but in Shinran-related texts through scripture titles, translator names, and fixed phrases.

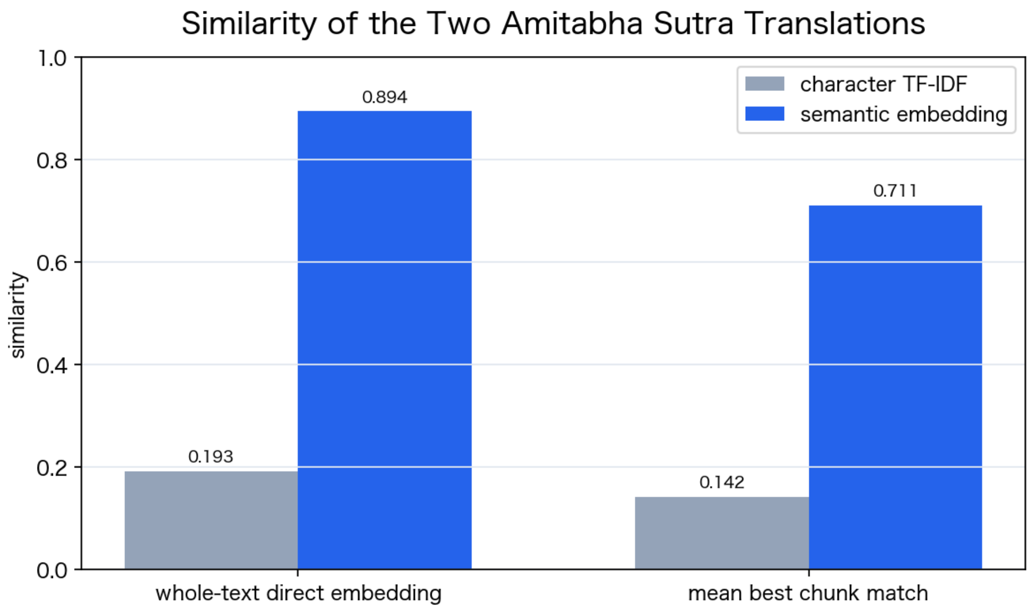

Figure9: Similarity comparison of the two Amitabha Sutra translations. Semantic embedding captures content proximity; character-level TF-IDF strongly reflects lexical and orthographic differences.

Table5: Metrics for the Two Amitabha Sutra Translations

Metric	Value	Description
Whole-text direct embedding similarity	0.8943	Value from an earlier small experiment embedding each translation as one input.
Chunk-mean vector similarity	0.8825	Main-analysis value from mean vectors of chunk embeddings.
Mean best chunk match	0.7108	Average proximity based on chunk-level correspondence.
Character TF-IDF whole-text similarity	0.1928	Lexical/orthographic similarity by character n-grams.
Character TF-IDF mean best chunk match	0.1424	Chunk-level lexical proximity.

Table6: Nearest Texts from the Amitabha Sutra

Neighbor text	Cosine similarity
稱讚淨土佛攝受經	0.8825
Larger Sutra	0.8694
Contemplation Sutra	0.8446
Lotus Sutra excerpt	0.8275
Kyogyoshinsho	0.8058

5.4 The Two Amitabha Translations in Shinran's Writings

Which Chinese Amitabha translation, or which commentarial tradition, Shinran relied on cannot be determined by semantic embeddings alone. Kumarajiva's and Xuanzang's versions are close in content, and embeddings strongly reflect that closeness. This paper therefore separates three kinds of evidence: explicit scripture and translator names, phrases that lean toward one translation, and semantic nearest-neighbor relations between *Kyogyoshinsho* chunks and the two translations.

Figure 10 shows source indicators in Shinran-related texts. On the examined *Nyushutsu Nimonge* page, *Praise of the Pure Land* and Xuanzang are explicitly named, and phrase indicators corresponding to Xuanzang's praise motif are also detected. In *Kyogyoshinsho*, phrases leaning toward Kumarajiva and references to Xuanzang's sutra title coexist. It is therefore better to read *Kyogyoshinsho* as a composite text in which the Three Pure Land Sutras, different translations, and commentarial traditions overlap, rather than assigning it to one translation alone.

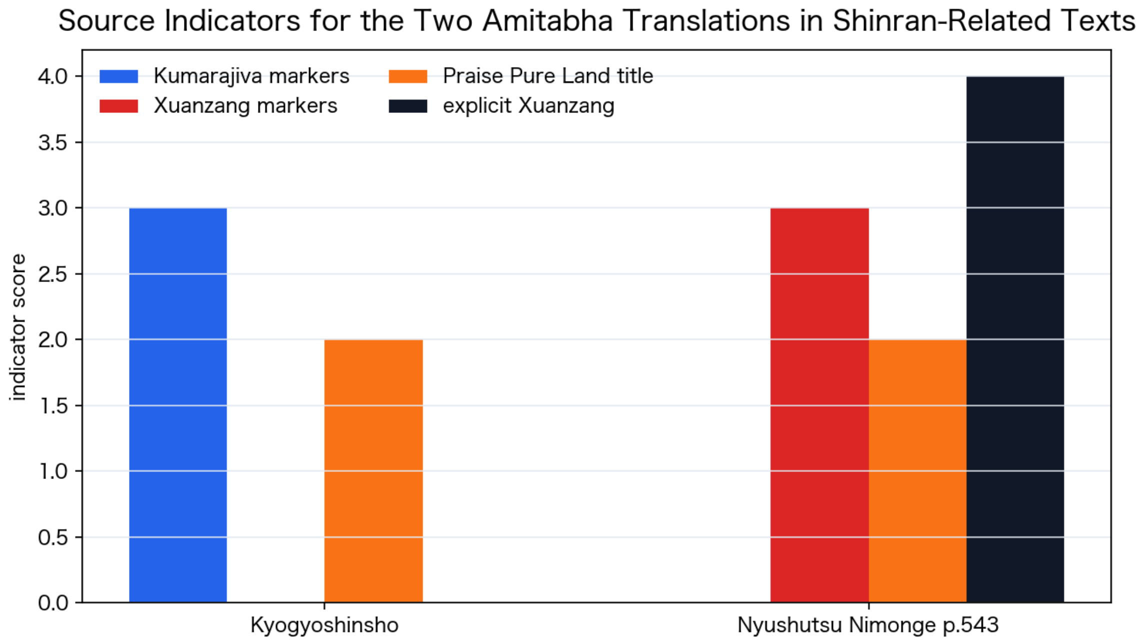

Figure10: Source indicators for the two Amitabha translations in Shinran-related texts. This is a small source/reference experiment separate from the global sectarian map.

The text-mean vector similarity from *Kyogyoshinsho* is 0.8058 to Kumarajiva's Amitabha Sutra and 0.8152 to Xuanzang's *Praise of the Pure Land*. Xuanzang's version is slightly higher, but the difference is only 0.0094. The chunk-level curve in Figure 11 also shows that proximity to the two translations alternates through the text, reflecting the mixture of sutra quotation, exegesis, founder citation, and doctrinal exposition.

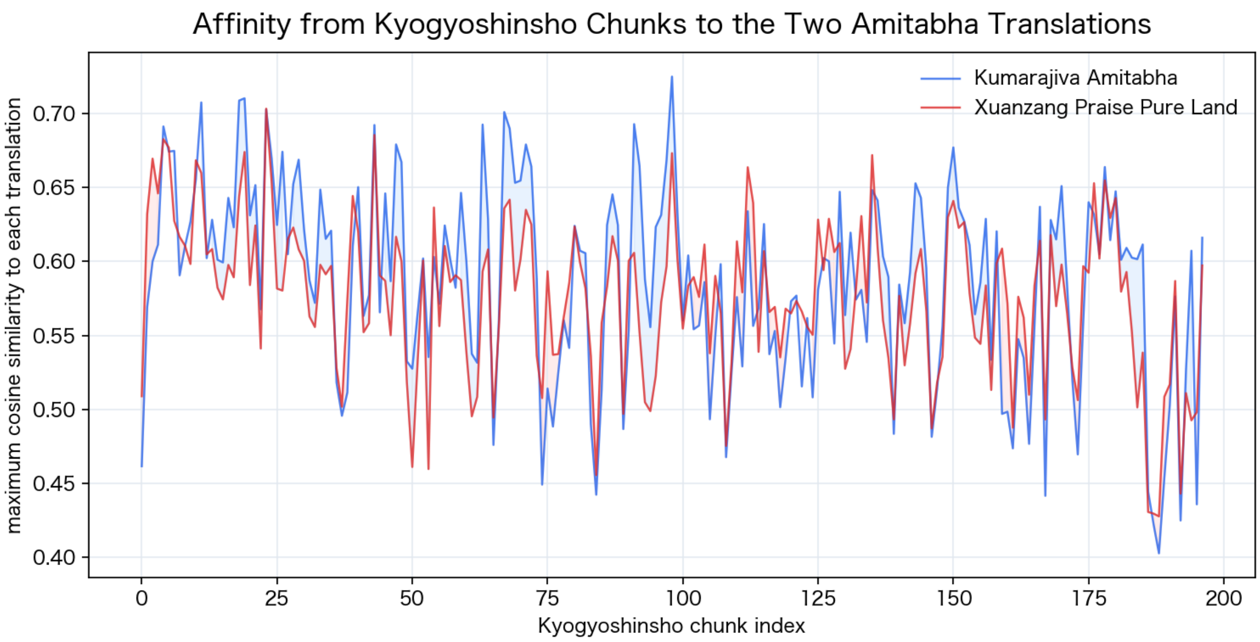

Figure11: Maximum similarity from Kyogyoshinsho chunks to the two Amitabha translations. Red indicates proximity to Xuanzang, blue to Kumarajiva.

Table7: Evidence for the Two Amitabha Translations from Shinran-Related Texts

Target	Kumarajiva side	Xuanzang side	Interpretation
Kyogyoshinsho	Characteristic phrases; chunk max 0.7250	Title reference; text mean 0.8152	Both translations and commentarial traditions overlap.
Nyushutsu Nimonge p.543	Not prominent	Explicitly names *Praise of the Pure Land* and Xuanzang	Strong evidence for Xuanzang reference.
Amitabha-kyo Shitchu	Not included	Not included	Priority material for the next stage.

The point of this small experiment is not to solve the two-translation problem by asking which translation is semantically closer. Embeddings capture the same-content relation of the two translations, but they blur differences in wording and citation paths. For Shinran-related source analysis, embedding neighbors should be used as an entry point, and scripture names, translator names, fixed phrases, and commentarial quotations should be overlaid as separate layers.

Figure 12 is the central trial result of this paper. It shows a three-layer source-mixture map for 197 chunks of *Kyogyoshinsho*, comparing them with the Larger Sutra, the Contemplation Sutra, Kumarajiva's Amitabha Sutra, and Xuanzang's *Praise of the Pure Land*. The same *Kyogyoshinsho* chunk sequence produces different source mixtures in the semantic, stylistic/lexical, and source-marker layers. Volume boundaries for the preface, Teaching, Practice, Faith, Realization, True Buddha and Land, and Transformed Buddha and Land fascicles are added from headings in the text. Because fixed-length chunks are used, boundary chunks are approximate and do not correspond to natural paragraph or fascicle units.

Table8: Mean Weights in the Kyogyoshinsho Three-Layer Source-Mixture Map

Layer	Unmarked	Larger Sutra	Contemplation	Kumarajiva Amitabha	Xuanzang Praise
Semantic	–	0.3705	0.2642	0.2135	0.1517
Stylistic/lexical	–	0.2942	0.2402	0.2369	0.2288
Source-marker	0.4958	0.4659	0.0303	0.0025	0.0054

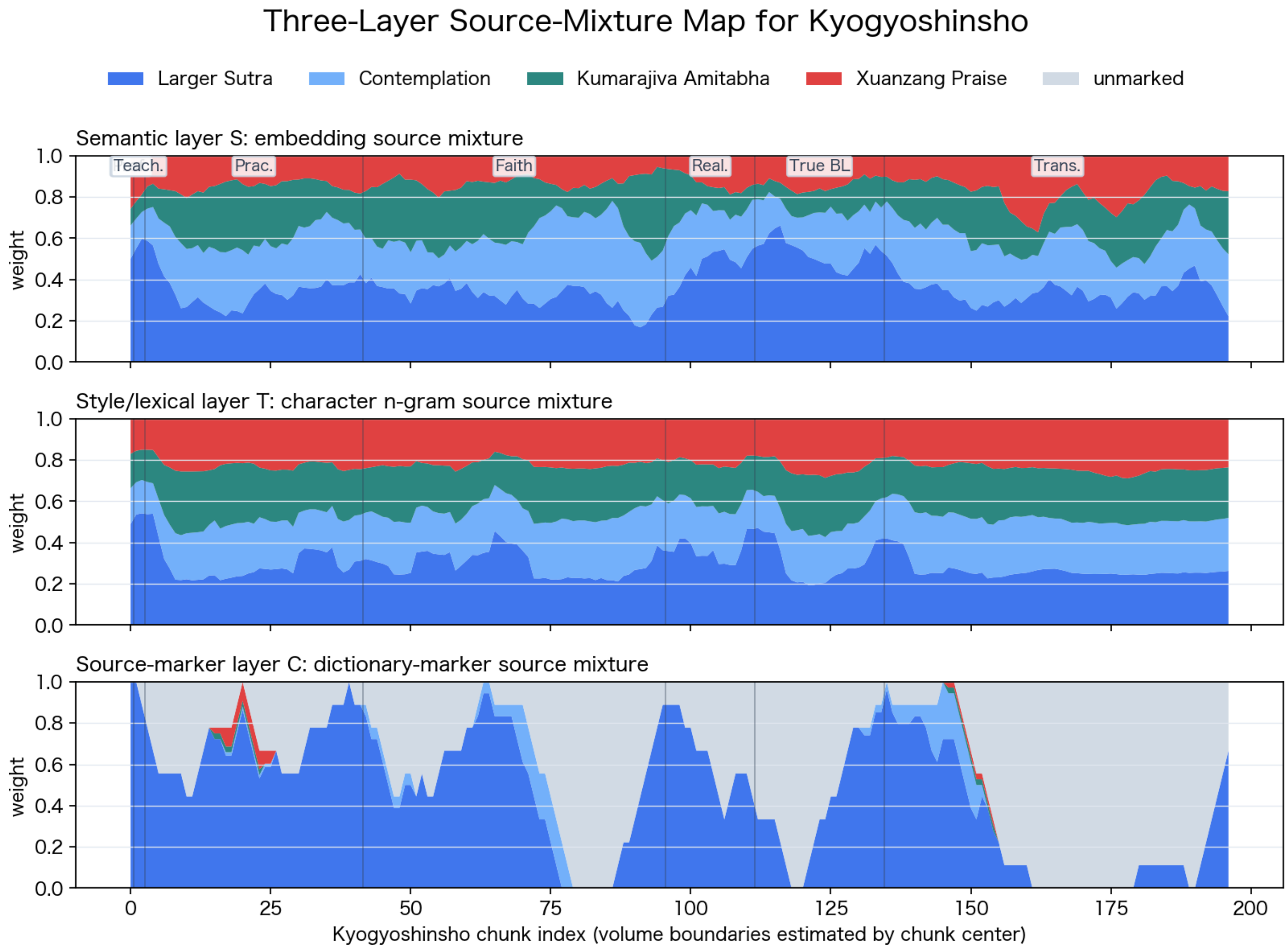

Figure12: Three-layer source-mixture map for Kyogyoshinsho. The upper panel is the semantic layer based on embedding similarity, the middle panel the stylistic/lexical layer based on character n-gram TF-IDF, and the lower panel the source-marker layer based on scripture-title, translator-name, and fixed-phrase dictionaries. Weights are relative within each layer. In the semantic and stylistic/lexical layers, all weight is necessarily distributed among the four reference sources; the largest weight is not therefore a true source attribution.

In the semantic layer, the Larger Sutra is the largest source in every volume. In the Teaching, Realization, and True Buddha and Land fascicles, its weight is especially high. In the Practice, Faith, and Transformed Buddha and Land fascicles, the Contemplation Sutra and the two Amitabha translations also occupy nontrivial portions. This should be compared with traditional readings of the volume structure; the computation alone does not determine the doctrinal character of each fascicle.

In the source-marker layer, the Teaching, Practice, and Realization fascicles are dominated by Larger Sutra markers, whereas the Faith, True Buddha and Land, and Transformed Buddha and Land fascicles are dominated by the unmarked category. This does not mean that those fascicles lack source relations. Rather, it suggests that many arguments, interpretations, and citation paths cannot be recovered by fixed phrases, scripture titles, and translator names alone.

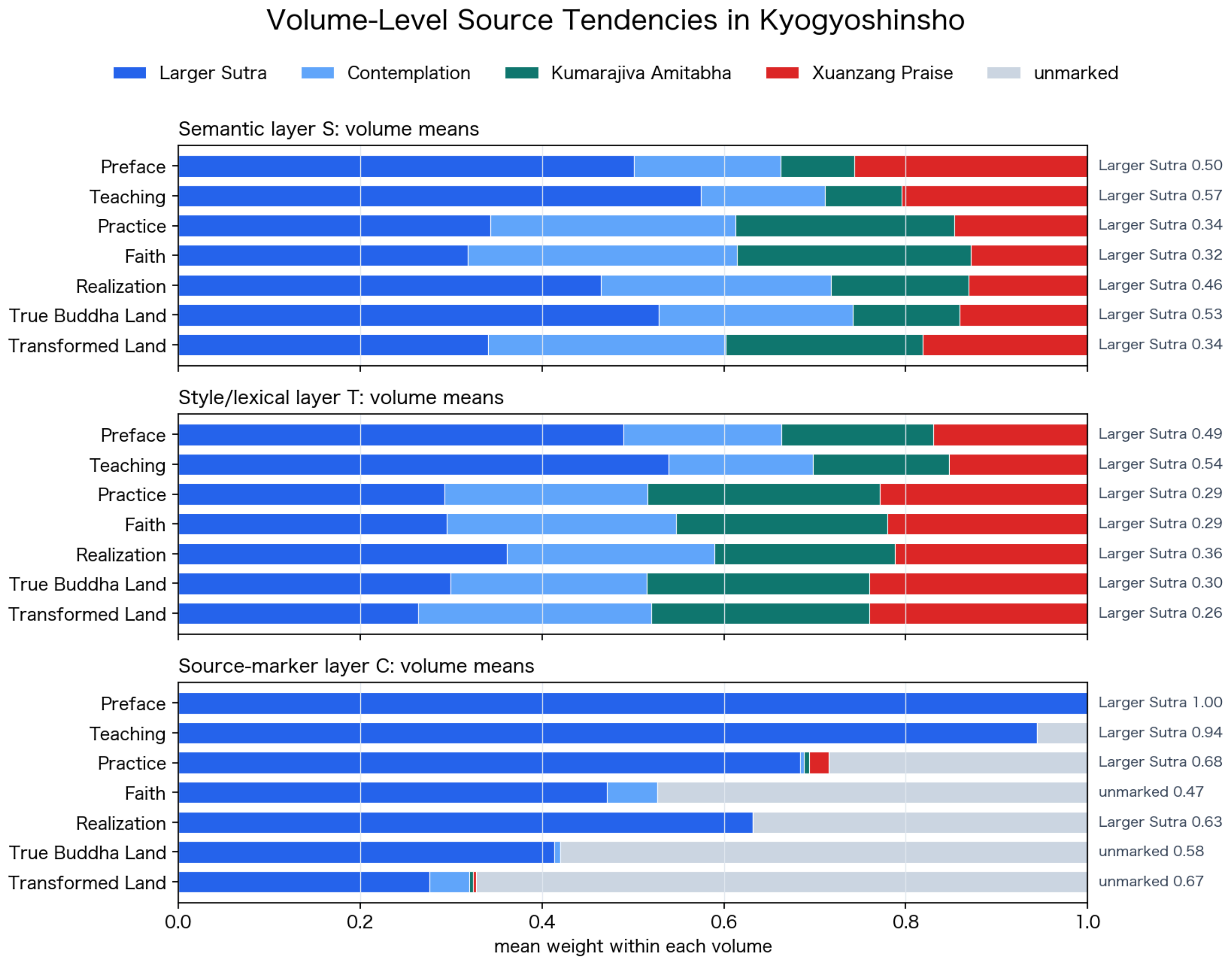

Figure13: Volume-level source tendencies in Kyogyoshinsho. Weights in the semantic, stylistic/lexical, and source-marker layers are averaged over chunks assigned to each volume.

Table9: Dominant Source by Volume-Level Mean in Kyogyoshinsho

Volume	Chunk range	Semantic max	Stylistic/lexical max	Source-marker max
Preface	0000–0000	Larger 0.5008	Larger 0.4899	Larger 1.0000
Teaching	0001–0002	Larger 0.5744	Larger 0.5390	Larger 0.9444
Practice	0003–0041	Larger 0.3436	Larger 0.2927	Larger 0.6840
Faith	0042–0095	Larger 0.3188	Larger 0.2950	Unmarked 0.4733
Realization	0096–0111	Larger 0.4648	Larger 0.3618	Larger 0.6319
True Buddha Land	0112–0134	Larger 0.5287	Larger 0.2992	Unmarked 0.5797
Transformed Land	0135–0196	Larger 0.3408	Larger 0.2640	Unmarked 0.6720

5.5 Translator Centroids

Translator centroids were created for Amoghavajra, Xuanzang, and Kumarajiva. The nearest texts to Xuanzang's *Praise of the Pure Land* were not other Xuanzang translations such as the Heart Sutra or *Samdhinirmocana-sutra*, but Kumarajiva's Amitabha Sutra and other Pure Land texts. In this small corpus, semantic proximity is controlled more strongly by theme and content than by translator. Translator-style detection will require more near-topic translation pairs and separate stylistic features.

Table10: Composition of Translator Centroids

Translator	Text count	Texts
Amoghavajra	2	Vajrasekhara, Rishukyo
Xuanzang	3	Praise Pure Land, Heart Sutra, Samdhinirmocana
Kumarajiva	5	Amitabha, Lotus, Kannon, Diamond, Vimalakirti

5.6 Multilingual Pilot

Using the 84000 English *The Teaching Explaining the Thought* (Toh 106) as the query text, cosine similarity was computed against Chinese texts. Table 11 shows the nearest neighbors.

Table11: Chinese Neighbors from the 84000 English Toh 106 Translation

Rank	Neighbor text	Cosine similarity
1	解深密經（T0676）	0.7662
2	維摩詰所説經（T0475）	0.6926
3	金剛般若波羅蜜經（T0235）	0.6922
4	般若波羅蜜多心經（T0251）	0.6167
5	佛説阿彌陀經（T0366）	0.5683

For the Toh 106/T0676 pair, the nearest Chinese text to the 84000 English translation is the corresponding Chinese *Samdhinirmocana-sutra*. This is not a general evaluation of multilingual performance; it is a small pilot suggesting that the three-layer mapping framework can be extended toward multilingual exploration. Vimalakirti and Diamond Sutra controls also appear in the 0.69 range, reflecting broader Mahayana thematic proximity. Future evaluation requires more corresponding and non-corresponding pairs and metrics such as top-1 accuracy, MRR, and ROC-AUC.

6 Discussion

The two Amitabha translations show that semantic embeddings can capture content similarity across lexical and translation differences. This is promising for detecting same-content or near-content Chinese translations. At the same time, character-level TF-IDF gives a low score for the same pair, because vocabulary, wording, proper-name forms, and orthography remain visible in character n-gram features.

This result connects embedding-based Buddhist studies and stylometric work on Chinese Buddhist translations. Semantic embeddings may extend the word- and concept-level exploration discussed by Huang and Wang[26] to texts and chunks. Translator and period differences, however, still require character features, particles, function words, PCA, and clustering of the kind used by Hung, Bingenheimer, and Wiles and by Bingenheimer, Hung, and Hsieh[24, 25].

Chunk distributions refine the interpretation of mean-vector proximity. The two Amitabha translations have close mean vectors and high chunk-neighbor mixing, suggesting strong local correspondence. By contrast, *Kyogyoshinsho* is close to the Three Pure Land Sutras in mean-vector space but less mixed at the chunk level, because it combines quotation, commentary, and doctrinal argument.

Pure Land and Shingon input groups show some thematic clustering, but this is not a classification of sects themselves. It is a visualization of the thematic proximity of selected reference texts. Founder writings such as *Kyogyoshinsho* may behave differently from single sutras because they synthesize many sources.

For Zen, scriptures alone did not separate sectarian traditions. This is better read as a data-layer problem than as a negative result about Zen. Comparing Zen traditions requires records of sayings, koan collections, monastic rules, and founder texts such as *Blue Cliff Record*, *Gateless Gate*, *Shobogenzo*, *Record of Linji*, and Obaku regulations.

Translator style should be studied with features separate from semantic vectors: character n-grams, particles and function words, sentence length, phrase structure, fixed translation equivalents, transcription versus semantic translation, and variation in Chinese renderings of the same Sanskrit terms.

The analysis therefore needs three layers. First is a semantic map for content, topic, and doctrinal proximity. Second is a stylistic/lexical map for wording, particles, phrase patterns, transcription terms, orthography, and sentence form. Third is a source-marker map for scripture names, translator names, fixed phrases, commentarial quotations, and founder-text references. The *Kyogyoshinsho* source-mixture map operationalizes this three-layer distinction.

The analysis of Amitabha sources in Shinran's writings illustrates the same problem. A text such as *Nyushutsu Nimonge*, which explicitly names Xuanzang's *Praise of the Pure Land*, provides strong evidence through names themselves. A composite work such as *Kyogyoshinsho*, however, cannot be decomposed automatically by semantic vectors alone. The three-layer map treats this limitation not simply as failure but as an interpretable difference among layers.

The source-marker map can connect with prior work on parallel-passage detection and intertextuality[27, 30]. This paper does not implement large-scale parallel-passage detection. Rather, it offers a way to identify which text relations should be examined more carefully with such tools.

The multilingual pilot suggests that semantic embeddings may recover same-text relations across languages. But the present experiment uses an English translation from 84000, not Tibetan text itself. Future work should jointly compare Tibetan, Chinese, and English in order to separate same-text identity, same-language effects, and translation style.

7 Limitations and Future Work

The corpus is small. Some large scriptures are not complete. Sectarian centroids are simple means. Translator centroids have few samples. PCA is only a visualization. Chunk ellipses are approximations in two-dimensional projection and should be supplemented with high-dimensional nearest-neighbor and distributional measures.

The three-layer source-mixture map is also preliminary. The stylistic/lexical layer is an initial character n-gram TF-IDF operationalization; it does not yet include particles, function words, sentence length, syntax, or fixed translation equivalents in a systematic way. The source-marker layer depends on a small dictionary, and an unmarked chunk does not mean absence of source relation. The semantic and stylistic/lexical layers distribute all weight among four sources and do not yet estimate an "other" category. The multilingual comparison is a one-English-text pilot; Tibetan text, chapter-level alignment, and multiple scripture families remain untested.

Future work should obtain stable full texts from SAT, add founder writings, records of sayings, ritual texts, and commentaries for each tradition, and build a separate stylistic vector space for translator, period, and Japanese kanbun features. Chunking should be compared between fixed windows and natural units such as fascicles, chapters, and paragraphs. Sectarian centroids should be weighted by scripture, liturgical, and founder-text status. The source-mixture map could be extended toward unbalanced optimal transport or chunk-transport maps with positional penalties for chapters and fascicles.

For Shinran's understanding of the Amitabha Sutra, the next priority is *Amidakyo Shitchu*（阿弥陀経集註）. It should not be embedded as a single undifferentiated text. Sutra text, interlinear notes, verso notes, and likely cited commentaries should be separated. Reference sources should include Kumarajiva, Xuanzang, Yuanzhao's *Amitabha Sutra Commentary*, Shandao-related texts, and Fazhao-related texts.

8 Conclusion

This preliminary experiment shows that semantic embeddings recover the proximity of same-content or near-content texts such as the two Amitabha Sutra translations and the thematic grouping of some Pure Land and Shingon scriptures. Representing texts as chunk distributions rather than only as mean points makes it possible to inspect internal spread and partial overlap. Combining mean-vector similarity and chunk-neighbor mixing distinguishes whole-text thematic proximity from local distributional overlap. In the Toh 106/T0676 pilot, the nearest Chinese text to the 84000 English translation is the corresponding Chinese text.

At the same time, semantic embeddings alone are insufficient for translator style and sectarian tradition. Translator habits require stylistic features such as character n-grams, translation-term correspondences, particles, and syntax. The present contribution is not a finished sect classifier but a basis for a three-layer exploratory map of Buddhist texts. Its distinctive point is to connect Japanese sectarian reference sets, the two Amitabha Sutra translations, Shinran's writings, chunk distributions, and a multilingual pilot within one framework, and to treat differences among semantic, stylistic/lexical, and source-marker layers as objects of interpretation.

Appendix: Terms, Models, and Tools

This appendix summarizes implementation terms needed to read the experiment. It describes how these terms are used here; tool specifications, availability, and pricing may change.

Corpus: A corpus is a collection of texts assembled for analysis. This paper uses small corpora of SAT Chinese Buddhist texts, Shinran writings from the Seikyo DB, and 84000 English translation material.

Preprocessing and normalization: Preprocessing includes extracting text from HTML and removing extra whitespace or interface text. Normalization aligns variant characters, punctuation, editorial marks, and spelling variants. This paper uses only basic preprocessing for semantic embeddings.

Token: A token is the unit used internally by a model or tokenizer. It is not identical to a character, word, or morpheme. Tokens are used here to create chunks of comparable length.

`tiktoken` and `cl100k_base`: `tiktoken` is OpenAI's Python tokenizer, and `cl100k_base` is one of the encodings used with OpenAI models[13]. It was used to split kanbun text into roughly 700-token chunks.

Chunks and overlap: A chunk is a segment of a longer text short enough for the embedding model. This experiment uses 700-token chunks with 100-token overlap.

Embeddings and `text-embedding-3-large`: An embedding is a high-dimensional vector representation of a text. This paper uses OpenAI's `text-embedding-3-large`[12]. Use requires an OpenAI API key and the Python OpenAI SDK.

Embedding space: The high-dimensional space in which text or chunk vectors are placed. Proximity in this space suggests semantic or thematic proximity, but it does not prove philological identity, quotation, or source relation.

Cosine similarity: A measure of whether two vectors point in similar directions. It is used here to compare embeddings.

Centroid: A working representative point produced by averaging multiple text vectors. A sectarian reference-group centroid is not the doctrinal center of a sect; it is the mean of selected reference texts.

PCA: Principal Component Analysis, used here to project high-dimensional vectors into two dimensions for visualization[19, 20].

TF-IDF and character n-grams: TF-IDF is a classical weighting method for text comparison[18]. Character n-grams are used here to inspect lexical and orthographic proximity separately from semantic proximity.

Stylometry: Statistical analysis of style through characters, words, particles, sentence length, syntax, and related features. It is treated here as a necessary layer beyond semantic embeddings.

Top-k neighbors and chunk-neighbor mixing rate: Top-k neighbors are the k most similar chunks to a given chunk. The mixing rate is the average proportion of top-k neighbors that belong to the other text when two texts' chunks are combined.

Three-layer source-mixture map: A visualization that shows how each target chunk relates to multiple reference sources as a mixture of weights, separated into semantic, stylistic/lexical, and source-marker layers.

Softmax: A function that converts multiple scores into weights summing to one. In this paper it converts source similarities into source-mixture weights.

Optimal transport and unbalanced optimal transport: Methods for mapping one distribution to another with minimal cost. They are future candidates for chunk-transport maps where some chunks may remain unmatched.

Top-1 accuracy, MRR, and ROC-AUC: Retrieval-evaluation metrics. Top-1 accuracy asks whether the correct item is ranked first. MRR measures how high the first correct item appears on average. ROC-AUC summarizes how well positive and negative pairs are separated across thresholds.

Parallel passages, intertextuality, and cross-linguistic semantic textual similarity: Parallel-passage detection looks for overlapping or corresponding passages. Intertextuality refers more broadly to textual relations such as quotation, allusion, reuse, and reference. Cross-linguistic semantic textual similarity compares meaning across languages.

Manifest, JSON, and HTML: A manifest is a structured list of target texts, metadata, and source information. JSON is a machine-readable data format used by the scripts and viewer. HTML is the document format used for the web paper, process report, and static viewer.

OpenAI API, API key, and SDK: The OpenAI API is the service interface used here to generate embeddings. An API key authorizes access to that service and must not be committed to git. The SDK is the Python library used by local scripts to call the API.

SAT, Seikyo DB, and 84000: SAT is the Taisho Shinshu Daizokyo Text Database; the Seikyo DB is the *Jodo Shinshu Seiten* database of the Jodo Shinshu Hongwanji-ha Research Institute; and 84000 provides English translations of Tibetan Buddhist texts. Source texts from these providers are not redistributed in the public data.

Author Responsibility and Scope of AI Assistance

The author of this paper is Daishin Ueyama. Because the project is also a prototype of AI-assisted humanities data-analysis paper production, the scope of AI assistance is stated explicitly. Recent publication ethics guidelines generally state that AI tools and large language models cannot be treated as authors because they cannot take responsibility for accuracy, integrity, originality, and final approval[14, 15, 16]. Contributor-role taxonomies such as CRediT are also widely used to clarify research contributions[17].

Daishin Ueyama was responsible for the overall conception, research questions, selection of sources, Buddhist-studies interpretation, reading of results, publication policy, and final content review. In CRediT terms, this corresponds mainly to Conceptualization, Resources, Supervision, Validation, Writing–review & editing, and Project administration. Final responsibility for the paper rests with Daishin Ueyama.

Codex GPT-5.5 xhigh, under Ueyama's instruction and review, supported implementation of scripts, text acquisition and preprocessing, the embedding pipeline, computation and organization of results, visualization, HTML/PDF generation, Git-based change recording, drafting, revision, and verification. In CRediT-like terms, this support corresponds to parts of Software, Data curation, Formal analysis, Visualization, Writing–original draft, and Writing–review & editing. Codex is not an author or co-author.

ChatGPT 5.5 Pro xhigh was used in the role of a reviewer, offering advice on problem framing, relation to prior work, originality, limitations, the explanation of the three-layer map, and wording for public release. These suggestions were selected and incorporated through Codex-assisted revision and Ueyama's judgment. AI-generated or AI-revised sentences, figures, code, and references remain materials requiring human verification; AI outputs are not treated as primary sources.

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During revision, ChatGPT 5.5 Pro xhigh was used in a reviewer role and provided useful comments on problem framing, relation to prior work, reproducibility, and limitations. The author gratefully acknowledges this assistance.

References

[1] SAT Daizokyo Text Database Committee, SAT Taisho Shinshu Daizokyo Text Database, https://21dzk.l.u-tokyo.ac.jp/SAT/ (accessed June 2, 2026)

[2] CBETA, CBETA Online, https://cbeta.org/ (accessed June 2, 2026)

[3] Buddhist Digital Resource Center, Buddhist Digital Resource Center, https://www.bdrc.io/ (accessed June 2, 2026)

[4] Jodo Shinshu Hongwanji-ha Research Institute, *Jodo Shinshu Seiten* Seikyo Database, https://j-soken.jp/ask/10831 (accessed June 2, 2026)

[5] Jodo Shinshu Hongwanji-ha Research Institute, Seikyo Database usage notes, https://j-soken.jp/ask/ask_5/640 (accessed June 2, 2026)

[6] Shinshu Seiten Search Site, *Nyushutsu Nimonge*, https://shinshuseiten.higashihonganji.or.jp/contents.html?id=2&page=543 (accessed June 2, 2026)

[7] Nagoya University Digital Archive, *Kan Muryoju-kyo Shitchu with Amidakyo Shitchu*, https://da.adm.thers.ac.jp/item/n004-20230901-00882 (accessed June 2, 2026)

[8] Ryusei Chiba, "On Quotations from Yuanzhao's *Amitabha Sutra Commentary* in Shinran's *Amidakyo*

Shitchu," 2014, https://buddhism.lib.ntu.edu.tw/jp/search/search_detail.jsp?seq=546211 (accessed June 2, 2026)
[9] Jodoshu, Entry on *Shosan Jodo Butsu Shoujukyo*, https://jodoshuzensho.jp/daijiten/index.php/%E7%A7%B0%E8%AE%83%E6%B5%84%E5%9C%9F%E4%BB%8F%E6%91%82%E5%8F%97%E7%B5%8C (accessed June 2, 2026)
[10] 84000: Translating the Words of the Buddha, *The Teaching Explaining the Thought*, Toh 106, https://read.84000.co/translation/toh106.html (accessed June 2, 2026)
[11] 84000: Translating the Words of the Buddha, Terms of Use, https://www.84000.co/documents/terms-of-use (accessed June 2, 2026)
[12] OpenAI, Embeddings, OpenAI Platform Documentation, https://platform.openai.com/docs/guides/embeddings (accessed June 2, 2026)
[13] OpenAI, tiktoken, https://github.com/openai/tiktoken (accessed June 2, 2026)
[14] International Committee of Medical Journal Editors, "Use of AI by Authors," https://www.icmje.org/recommendations/browse/artificial-intelligence/ai-use-by-authors.html (accessed June 2, 2026)
[15] Elsevier, "The use of AI and AI-assisted technologies in writing for Elsevier," https://www.elsevier.com/about/policies-and-standards/the-use-of-generative-ai-and-ai-assisted-technologies-in-writing-for-elsevier (accessed June 2, 2026)
[16] Springer Nature, "Artificial Intelligence (AI)," https://link.springer.com/brands/springer/journal-policies (accessed June 2, 2026)
[17] National Information Standards Organization, "CRediT: Contributor Role Taxonomy," https://credit.niso.org/ (accessed June 2, 2026)
[18] Gerard Salton and Christopher Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, 24(5):513–523, 1988
[19] I. T. Jolliffe, *Principal Component Analysis*, Springer, 2nd edition, 2002
[20] Fabian Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, 12:2825–2830, 2011
[21] John F. Burrows, "Delta": a Measure of Stylistic Difference and a Guide to Likely Authorship, *Literary and Linguistic Computing*, 17(3):267–287, 2002, https://doi.org/10.1093/llc/17.3.267
[22] Moshe Koppel, Jonathan Schler, and Shlomo Argamon, "Computational Methods in Authorship Attribution," *Journal of the American Society for Information Science and Technology*, 60(1):9–26, 2009, https://doi.org/10.1002/asi.20961
[23] Seishi Karashima, *A Glossary of Kumarajiva's Translation of the Lotus Sutra*, The International Research Institute for Advanced Buddhology, Soka University, 2001, https://glossaries.dila.edu.tw/glossaries/KKJ?locale=en (accessed June 2, 2026)
[24] Jen-Jou Hung, Marcus Bingenheimer, and Simon Wiles, "Quantitative Evidence for a Hypothesis regarding the Attribution of Early Buddhist Translations," *Literary and Linguistic Computing*, 25(1):119–134, 2010, https://doi.org/10.1093/llc/fqp036
[25] Marcus Bingenheimer, Jen-Jou Hung, and Cheng-en Hsieh, "Stylometric Analysis of Chinese Buddhist Texts: Do Different Chinese Translations of the Gandavyuha Reflect Stylistic Features that are Typical for their Age?" *Journal of the Japanese Association for Digital Humanities*, 2(1):1–30, 2017, https://doi.org/10.17928/jjadh.2.1_1
[26] Shu-Ling Huang and Yu-Chun Wang, "Word Embedding in Buddhist Studies: On the Basis of

Evaluation of Word Embedding Models," *Journal of Digital Archives and Digital Humanities*, 12:43–82, 2023, https://doi.org/10.6853/DADH.202310_(12).0003

[27] Sebastian Nehrdich, "A Method for the Calculation of Parallel Passages for Buddhist Chinese Sources Based on Million-scale Nearest Neighbor Search," *Journal of the Japanese Association for Digital Humanities*, 5(2):132–153, 2020, https://doi.org/10.17928/jjadh.5.2_132

[28] Rafal Felbur, Marieke Meelen, and Paul Vierthaler, "Crosslinguistic Semantic Textual Similarity of Buddhist Chinese and Classical Tibetan," *Journal of Open Humanities Data*, 8:23, 2022, https://doi.org/10.5334/johd.86

[29] Ligeia Lugli, Matej Martinc, Andraz Pelicon, and Senja Pollak, "Embeddings Models for Buddhist Sanskrit," In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 3861–3871, 2022, https://aclanthology.org/2022.lrec-1.411/

[30] Dharmamitra, DharmaNexus, https://dharmamitra.org/nexus (accessed June 2, 2026)

[31] Sebastian Nehrdich et al., "MITRA: A Large-Scale Parallel Corpus and Multilingual Pretrained Language Model for Machine Translation and Semantic Retrieval for Pali, Sanskrit, Buddhist Chinese, and Tibetan," arXiv preprint arXiv:2601.06400, 2026, https://arxiv.org/abs/2601.06400